

Presenter Reference Notes – September 2026

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Slide 1: NDNF: A Data-Centric Service Framework for

Secure, Collaborative, and Dynamic Networked Applications Department of Computer Science University of Memphis Advisor: Dr. Lan Wang PhD Proposal Defense September 2026

Slide 2: Service Discovery: Several Candidates, One Executor

The User knows the required service. Reachability, capability and willingness may change. Eligible Providers return ACK Data. The User selects one Provider to execute. Candidate Providers A, B, C User selects B B executes the service Initial NDNF: discover alternatives and select one Provider per invocation, not multiple collaborating roles. Arrows show the selection process.

Slide 3: From Invocation to Collaboration: Two Research Scopes

Foundation: evaluate discovery and authorization The initial prototype selects one Provider. Discovery confidentiality, service permissions and management costs matter even without collaboration. Extension: coordinate complementary roles Some tasks need several Providers with different responsibilities. Separate calls do not define a shared role assignment or enforce dependencies between them. Evaluate the invocation foundation; develop and validate the collaboration extension. Existing invocation results do not establish collaboration correctness.

Slide 4: What NDN Provides—and What It Does Not

NDN building block Remaining application responsibility Named Data and signatures [1] Validate signatures and signing-key relationships under trust rules; check task permissions separately Interest retrieval and caching [1,26] Enforce task identity, dependencies, and deadlines; caching is not persistent storage Dataset synchronization / SVS [2,3] Discover new names, then retrieve Data; define service-message state transitions NAC [4] / enhanced NAC-ABE [19] Use content encryption for read access; separately check service-invocation permissions and state NAC: Name-based Access Control; ABE: attribute-based encryption; SVS: State Vector Sync. NDNF builds on these mechanisms; it does not invent them.

Slide 5: Three Research Questions

RQ1: How can NDNF discover willing Providers, keep descriptors confidential from unauthorized readers, and check service permissions before execution, and at what cost? RQ2: How can single-Provider invocation be extended to roles and named dependencies across Providers, whether Data comes from producers or caches? RQ3: When do these mechanisms improve completion, latency, or execution efficiency, and when do their costs outweigh the benefits? RQ1 evaluates the invocation foundation; RQ2 studies its collaboration extension; RQ3 measures both scopes separately.

Slide 6: Related Work Already Provides Important Pieces

Prior work What the comparison must acknowledge NFN [6], NFaaS [7] Named computation and function execution/placement RICE [9], NSC [10] Invocation and security mechanisms; NSC validates signed calls using application trust rules DNMP [12] Role- and command-constrained signing and verification CaSCON [15] Service-composition research; requires a feature-level comparison before claiming a new composition mechanism gRPC [17,27] Complete configurations can include discovery/resolver integration, not only fixed addresses Collaboration research: make runtime assignments constrain intermediate Data use. Discovery and authorization are evaluated separately.

Slide 7: Contributions and Evidence Are Different Things

Intended contribution Current evidence boundary Discovery / authorization design and tradeoffs Invocation implementation and bounded experiments exist; full alternative lifecycle-cost comparison remains open ACK-derived plan and local dependency checks Design and implementation work; direct/cache and failure validation remains incomplete Cross-case empirical findings Existing mobility/work-efficiency and small-model results support limited configurations, not all new collaboration claims Proposal status, September 2026: implemented, locally tested, measured, and fully qualified are not interchangeable. [28]

Slide 8: Participants and Trust Bootstrap

User: requests a service and selects a Provider; the extension also assigns collaboration roles. Provider: offers and executes a service. A node may play both roles. Controller: administers identities and service permissions approved by the organization. Participants use provisioned trust anchors and validation rules for certificates and Controller material. Unknown runtime candidates are not unregistered authorization identities. The User discovers current participation; the Controller does not choose this task's Providers.

Slide 9: Single-Provider Invocation: Four Named Data Types

Data Logical purpose Request Describe required service and participation constraints; include protected discovery information and Requester Challenge ACK Report positive or negative participation; positive ACK supplies selection information and Provider Challenge Selection Assign selected execution; provide input/references, challenge response and protected request-scoped keys Response Return status, results or result references; the User still validates them before use Sync helps discover names; Interests retrieve Data. The Provider checks Selection before execution; a positive ACK alone does not start the task.

Slide 10: Request: Discovery Descriptor; Selection: Task Input

Discovery descriptor Needed to decide whether to participate: area, deadline, sensor/model and resource requirements. Application input Needed to execute the selected task: images, detailed processing parameters, or protected input references. The boundary follows necessity. A location needed to assess coverage belongs in the protected descriptor; an image needed only for processing can wait.

Slide 11: Reason 1: Confidential Discovery Before Selection

The User knows the service, but not yet the currently willing Provider identities. An observation area and deadline can reveal intended activity even without the full application input. The User encrypts the descriptor using ABE and the service attribute, without listing Provider identities. [19] Confidential

from parties lacking the required decryption capability. Authorized but unselected Providers can read it; names and traffic patterns remain visible.

Slide 12: Reason 2: Reuse Service-Permission Management

KP-ABE encodes an access policy in each entity's decryption key. [19] NDNFSF combines service entitlements in that policy while retaining identity-signing credentials. [28] Challenge checks reuse these permissions for invocation; User permission keys and challenge processing still add cost. This avoids a separate role-based invocation policy in NDNFSF. It does not eliminate key updates or prove lower total cost; role-based designs can also reuse credentials.

Slide 13: How NDNFSF Checks Service Permissions

1. The User places a Requester Challenge in ABE-protected Request content. 2. The Provider returns it in ACK Data; the User checks the value against the active request. 3. The Provider encrypts its challenge in ACK content. The User returns the value in Selection; the Provider checks it before execution. Both sides also validate signatures under trust rules and check service permissions and request state. Challenge responses show access to protected values; they do not grant permissions or prevent deliberate sharing.

Slide 14: DNMP: Trust-Schema Command Authorization

Distributed Network Measurement Protocol (DNMP) [12] authorizes measurement commands: 1. The Network Key anchors role keys, device keys, and the schema. 2. The client uses target, probe type, arguments, and user role to find an allowed signing key; among matching keys, it chooses the least privileged. 3. The receiving Network Observer Daemon validates the publication under the schema before dispatching the probe. DNMP also uses publication lifetimes and retained state against replay, assuming loosely synchronized clocks. These are concrete command-authorization mechanisms, not just identity checks.

Slide 15: DNMP-inspired Authorization versus NDNFSF

Design Execution authorization Discovery confidentiality DNMP-inspired [12] Check role-constrained signatures and NDNFSF invocation state Pair with service-group encryption or ABE NDNFSF ABE-backed User/Provider check signatures, challenge responses, service permissions and request state User encrypts descriptor content with service-level ABE Proposed comparison: retain NDNFSF discovery and Selection, with input/result protection and revocation checks in both designs. Trust Schema [16] alone does not encrypt Content; adapting DNMP authorization is not an original DNMP feature. No measured cost advantage is claimed.

Slide 16: When Another Authorization Design May Be Better

Known Providers and simple rules: signed invocation plus recipient encryption may be sufficient. A small, stable receiver group: ordinary group-key management may be attractive. Rich command-argument restrictions: DNMP-style rules make those constraints explicit. [12] Many changing entitlements: measure both policy/key updates and credential/rule updates before claiming a management advantage. Compare complete combinations of authorization and encryption. ABE-backed authorization is one design choice, not a universally superior solution.

Slide 17: Authorization Lifecycle and Costs

The Controller publishes signed permission updates; participants apply them only after retrieval and validation. Request-scoped keys protect input/results. Selection delivers keys to selected Providers; the requesting User retains access. Status checks reject withdrawn permissions; key rotation restricts future encrypted exchanges. Current ABE rotation is Controller-wide. [28] Revocation has costs: measure update propagation, rejection delay and rekey traffic. Disconnected nodes cannot learn updates immediately; previously disclosed keys and plaintext cannot be recalled.

Slide 18: Threat Model: What Is Trusted?

Trusted: Controller authorization and the User's planning decision. Untrusted delivery: an object may arrive through a producer, forwarder, cache, or repository and still requires validation. Adversarial cases to test: wrong identities, permissions, challenges, request state, roles, dependencies, or execution identities. Not established by signatures/ABE: honest resource reports, correct inference, truthful images, or resistance to authorized key sharing. No Byzantine-coordinator consensus claim; no traffic-analysis or instantaneous-revocation guarantee.

Slide 19: RQ2 Extension: From Positive ACKs to a Role Plan

1. Application supplies roles, dependency template, constraints and deadline; the template may be handwritten.
2. The User closes the ACK window and freezes the eligible, validated positive ACKs.
3. The User applies a replaceable planning policy and signs the resulting role assignments and dependencies.
4. Each Provider validates the plan, derives permitted inputs and rechecks its resources before starting. The snapshot freezes observed candidates for this decision, not every Provider in the network or their future availability.

Slide 20: A Plan Is More Than a Request ID

Existing request ID Proposed task-specific plan rules Identifies the request; requires a validated message
Assigns exact Providers to collection and processing roles Does not alone specify predecessors Declares which producer role may supply each named dependency Does not alone describe key scope Associates permitted objects, execution identity, and authorization/key scope Example: within the same request, a service-authorized UAV that was not assigned "view 1" must not satisfy the view-1 dependency.

Slide 21: Plan-to-Data Enforcement: Check Before Advancing

1. Validate the Data signature using the appropriate trust rules.
2. Check active request, plan, execution identity and deadline.
3. Verify that the signer is the assigned predecessor Provider.
4. Check that the object matches the declared edge, permitted name, data format and key scope.
5. Require complete validated input before the dependent computation starts. Apply the same rules to direct, cached and repository-fetched Data. This is a proposed acceptance contract; full end-to-end qualification remains required.

Slide 22: Collaboration Recovery Needs Explicit State

Event Required behavior Late ACK Do not silently change the committed plan Duplicate Selection Follow recorded one-time state; do not implicitly repeat a side effect Incomplete dependency Wait within

policy/deadline, recover, or fail; do not execute on partial data Provider replacement Authorize a new assignment; reject old execution actions and specify permitted result reuse Cancellation Prevent late messages from reauthorizing cancelled work Network re-retrieval is not task re-execution. Compensation is not an atomic transaction or exactly-once physical action.

Slide 23: Proposed UAV Collaboration: Capture, Analyze and Return

An operator requests ground-area views from unmanned aerial vehicles (UAVs), followed by analysis. 1. Sensing UAVs sign encrypted ViewImage Data under their authorized namespaces. 2. The compute Provider retrieves images over assigned edges and publishes Annotation Data. 3. Its encrypted result gives the requester original image names and permitted image keys. 4. The requester retrieves the original images and displays the annotations. Original image Data remains under its producer's name. GPS alone does not establish camera visibility.

Slide 24: What the UAV Case Must Demonstrate

A complete authorized collection–processing–return task before the deadline. Correct handling of unavailable views, delayed images and a failed processing Provider. Reject wrong-role/edge inputs; accept cached images only when they satisfy the same task checks. MiniNDN [22] and PX4 SITL [23] results clearly distinguished from physical-flight evidence. Higher recognition/localization accuracy is a separate claim requiring ground truth and a single-view control. It is not assumed here.

Slide 25: Distributed Inference: Named Computation Dependencies

Provider A layer group 1 Provider B layer group 2 Provider C output stage activation activation Pipeline: sequential layers/groups; tensor partitioning additionally needs supported collectives. ONNX [25] describes the graph; export alone does not implement distributed execution. Repo [24] stores artifacts; forwarding caches do not guarantee persistence. The framework contribution is checked runtime assignment and data exchange, not model partitioning itself.

Slide 26: DI State and Evidence Boundaries

Model weights, reusable artifacts, request activations and KV state have different lifetimes. With KV caching, the runtime reuses prior key/value (K/V) tensors and supplies next-step inputs; resident weights do not replace this state. [31] An in-process test does not establish correct execution across processes or devices. Freeze one supported model/partition first; tensor/hybrid and LLM streaming need their own evidence if claimed. Earlier small-model measurements do not validate the newer distributed inference and recovery paths. [28]

Slide 27: RQ3: Define the Baselines Before Comparing

NSC [10]: named service calls; describe the tested executor, timeout, queue and recovery behavior. gRPC [17]: describe resolver, sequential/parallel policy, deadlines, retry and cancellation. Match Provider count, capacity, task, topology/traces, publication times and measurement windows. Report fixed-list and dynamically discovered configurations as different configurations. An incomplete baseline cannot establish a limitation of the full framework. Historical unmatched data remain diagnostics.

Slide 28: Mobility: Completion under Different Coverage Conditions

Ten seeds (62–71), four Providers/system, 2 m/s; mean seed completion rate. [28] AP radius NDNFSF gRPC NSC 50 m 54.57% 53.80% 55.47% 100 m 97.97% 97.97% 98.33% 150 m 100.00% 100.00% 100.00% No overall NDNFSF advantage is established. The next slide uses a separate holdout campaign (seeds 72–81), not a subset of these runs.

Slide 29: Mobility: Lower Tail Latency in a Defined Subset

Seeds 72–81; four Providers/system; 100 m, 2 m/s; 1,312 aligned switching requests. [28] Comparison Seed-p95 difference 95% CI NDNFSF minus gRPC-SEQ-4 -597 ms [-992, -200] ms NDNFSF minus NSC-4 -2618 ms [-2916, -2315] ms Completed: NDNFSF 1,311; gRPC 1,297; NSC 1,296 (of 1,312 each). Mean paired seed-p95 differences; each p95 uses that system’s successful requests. Different failures limit the comparison; no general advantage is inferred.

Slide 30: Fewer Execution Starts than the Parallel Baseline

Six-seed, capacity-matched comparison; 1,800 requests per system. [28] Configuration Completed requests Execution starts NDNFSF 1,798 / 1,800 1,800 gRPC-PAR-4 1,800 / 1,800 7,180 NSC-4 also completed 1,800/1,800, with 1.002 starts/request (rounded). No work-efficiency advantage over NSC is claimed. Fewer starts than gRPC-PAR-4 is not proof of fewer bytes, less CPU time or lower energy. Similar observed completion is not proven equivalence.

Slide 31: A Useful Negative Result: Splitting Can Be Slower

Earlier small Qwen-style ONNX model; ten successful runs per layout. [28] Layout Mean latency p95 Shared-backbone distributed layout 259.4 ms 283.9 ms Single Provider 186.8 ms 206.3 ms Do not split merely because several Providers exist. Communication/setup overhead is a candidate explanation, not a causal result established by this measurement. Retained historical diagnostic; raw-run/model provenance remains to be rechecked. Not evidence for the newer distributed design.

Slide 32: Remaining Experiments Follow the Claims

RQ Required evidence Outcome and cost RQ1 Full authorization alternatives; unauthorized call, challenge/replay and withdrawal cases No unauthorized handler start; discovery delay, key/policy size, status delay and rekey traffic RQ2 One UAV and one DI graph; wrong role/edge; direct/cache parity; cancellation/replacement Correct task completion; validation, transfer and recovery cost RQ3 Matched workload and failure schedules; feasible single-device control; cold/warm state Completion plus latency, redundant work, bytes, uncertainty, and cases with no advantage Planned tests are not completed findings. Record all scheduled runs and failure boundaries.

Slide 33: A Bounded Path to Spring 2027

Internal target Required deliverable September 2026 Agree scope/RQs; freeze one UAV graph and one DI partition; resolve correctness blockers October–November 2026 Complete registered experiments; freeze evidence and write bounded RQ answers December 2026 Complete integrated advisor-review draft January 2027 Committee review and readiness decision; reduce scope or revise timing if core evidence is missing February 2027 Target defense, conditional on evidence and committee approval Committee draft:

two weeks before defense. Spring 2027 application/candidacy: February 5; final ProQuest: March 26. Recheck before filing. [29,30]

Slide 34: Expected Research Outcomes

RQ1 answer: when ABE-encrypted discovery and ABE-backed service authorization justify their management and revocation costs. RQ2 answer: exactly which task-assignment mistakes are rejected and which recovery behavior is supported. RQ3 answer: measured benefits, negative results and deployment limits. Reproducible configurations and evidence linking each conclusion to its tested scope. The outcome is a tested service protocol and an evidence-backed account of when its mechanisms are worth their cost.

Slide 35: Discussion: Is This Scope Sufficient and Feasible?

1. Are the three research questions specific enough to support defensible answers? 2. Are the authorization alternatives complete and fairly configured? 3. Are one UAV graph and one supported DI partition sufficient to test the abstraction? 4. Which optional features should be excluded if they delay the core evidence? NDN-SF: evaluate runtime discovery and authorized invocation; extend the foundation to role-aware collaboration.

Slide 36: Backup: Discovery Configuration Is Not Framework Capability

Provider-transition configuration Completed NDN-SF, no pre-registered Provider addresses 357 / 357 Static-list NSC/gRPC configurations 0 / 357 Pre-registered baseline variants 357 / 357 This isolates the configured discovery boundary. Resolver/discovery integration can update candidates; it is not proof that gRPC or NSC cannot do so. [27,28]

Slide 37: Backup: Historical Loss Evidence Is Not a New Matched Test

Historical per-link loss NDN-SF completed requests 1% 599 / 600 3% 598 / 600 10% (best recorded run) 539 / 600 A selected run is not an aggregate estimate. Do not combine these numbers with another batch and label the result “matched.” [28] Per-link packet loss is not the probability of service failure: retrieval retries and the request deadline also affect completion.

Slide 38: References: NDN and Authorization

[1] L. Zhang et al., “Named Data Networking,” ACM SIGCOMM CCR, 2014. [2] T. Li et al., “A Brief Introduction to NDN Dataset Synchronization (NDN Sync),” MILCOM, 2018. [3] P. Moll et al., “A Brief Introduction to State Vector Sync,” NDN-0073, rev. 2, 2021. [4] Z. Zhang et al., “NAC: Automating Access Control via Named Data,” MILCOM, 2018. [12] K. Nichols, “Lessons Learned Building a Secure Network Measurement Framework using Basic NDN,” ACM ICN, 2019. [16] Y. Yu et al., “Schematizing Trust in Named Data Networking,” ACM ICN, 2015. [19] S. Dulal et al., “Enhancing NAC-ABE to Support Access Control for mHealth Applications and Beyond,” arXiv:2311.07299, 2023. [26] NDN Project, “Data Packet,” NDN Packet Format Specification. docs.named-data.net/NDN-packet-spec/current/data.html (accessed Sept. 2026).

Slide 39: References: Invocation and Composition

[6] C. Tschudin and M. Sifalakis, “Named Functions and Cached Computations,” IEEE CCNC, 2014. [7] M. Krol and I. Psaras, “NFaaS: Named Function as a Service,” ACM ICN, 2017. [9] M. Krol et al., “RICE: Remote Method Invocation in ICN,” ACM ICN, 2018. [10] D. Meirovitch and L. Zhang, “NSC—Named Service Calls, or a Remote Procedure Call for NDN,” NDN-0074, rev. 1, 2021. [15] CaSCON, SMARTCOMP 2025. Primary publication record: RIT, “Barrios presents at IEEE SmartComp.” Full feature-level comparison remains a research obligation. [17] gRPC Authors, “Core Concepts, Architecture and Lifecycle,” gRPC documentation. [27] gRPC Authors, “Custom Name Resolution,” grpc.io/docs/guides/custom-name-resolution/ (accessed Sept. 2026).

Slide 40: References: Tools, Evidence and Degree Planning

[22] Mini-NDN, project documentation: minindn.ndn.today. [23] PX4, “Software-in-the-Loop Simulation,” PX4 User Guide. [24] NDN Project, “Repo Protocol Specification,” [repo-ng](https://repo-ng.org/) documentation. [25] ONNX, “Open Neural Network Exchange Intermediate Representation Specification.” [28] NDN NSF, [research-revision-audit.md](#), companion claim/evidence ledger, September 9, 2026. Contains campaign paths and unresolved acceptance gates. [29] University of Memphis CS, “Information for Computer Science Ph.D. Students,” [forms_grad_phd_programinfo.pdf](#) (accessed September 9, 2026). [30] University of Memphis Graduate School, “Graduation Information,” Spring 2027 calendar (accessed September 9, 2026). [31] ONNX Runtime, “How to configure the past present share buffer,” GenAI documentation (accessed September 11, 2026).